

1. Student Data Base Management using double linked list

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* Title :

Student Database management using double linked list

* Problem statement :

Design and implement a console-based student data base management system in Cpp using double linked list

The system should store and manage student information such as roll number, Name, branch and marks.

The user can perform different operations such as adding a student, displaying all student, searching for a student, updating student details and deleting a student record

The double linked list allows traversal in both forward and backward direction making student record management easier and efficient.

* Objectives

- To implement a student data base management system using c++ programming.
- To understand and implement the concept of double linked list.
- To understand the use of dynamic memory allocation.
- To perform operations such as insertion deleting and searching.
- To implement forward and backward Traversal of double linked list.
- To implement student record searching and updating.
- To develop a simple menu driven application using c++.
- To understand how linked lists can be used for managing dynamic data.

Algorithm

- step 1 :- start the program
- step 2 :- Initialize the number of records as 0.
- step 3 :- Display the menu to the user.
- step 4 :- Ask the user to select an option
- step 5 :- If the option is add, take the student details and store.
- step 6 :- Show all the stored records
- step 7 :- If the option is search, ask for all number and display the student
- step 8 :- If the option is update ask for roll number, and new details then update the record.
- step 9 :- If the option is delete, ask for all roll number
- step 10 :- If the option is exit, terminate the program

* Flowchart

